

II. M.Sc., ZOOLOGY

Elective paper IV. Applied Entomology

Dr. M.MUTHUSAMY. Associate professor

UNIT:III

PEST OF PADDY

Rice, *Oryza sativa* is the most important and staple food crop for more than two third of the population of India and more than 65 % of the world population.

More than 100 Insect species are associated with rice crop at one stage or the other stage

20 of these are pests of major economic significance and these pests are classified as

Borer pests

Sucking pests

Defoliator pests

Non Insect pests

Insect PESTS OF PADDY

Borer pests of paddy



Yellow stem borer
Scirpophaga incertulas



Gall midge or Gall fly
Orseolia oryzae

Sucking pests of paddy



Green leaf hoppers (GLH)
1. *Nephotettix nigropictus*
2. *N. virescens*



White leaf hopper (WLH)
Cofana spectra



Brown Plant hopper (BPH)
Nilaparvatha lugens



Earhead bug
Leptocorisa oratoria



Thrips
Stenchaetothrips biformis



Mealybugs
Brevinnia rehi

Paddy stem borer : *Scirpophaga incertulas* (Walker) (Pyralidae : Lepidoptera)

- ✧ The adults are yellowish white with orange yellow front wings and they have sexual dimorphism
- ✧ The female moth is bigger than the male and has a centrally situated black spot on each of the fore wings
- ✧ The females also have a prominent tuft of brownish silken hair at the tip of their abdomen
- ✧ Moths are attracted to light in the night during March-April and October-November



Yellow Stem Borer/Adult



Male



Female

Male moths are small with pale yellow fore wings and without black spot in the centre and abdomen slender with out tuft of hairs

Female moths are bigger fore wings are bright yellow and a prominent black spot in the centre of each and the anal end with tuft of yellowish hairs



Yellow Stem Borer

Egg Mass

Eggs are creamy coloured , flat, oval, laid in masses and covered with buff coloured hairs on the tip of the leaf



Yellow Stem Borer

Larva

Body: cream

Head: reddish-brown

The body of the larva is cream or Pale white or yellow colored and the head is reddish brown



Yellow Stem Borer

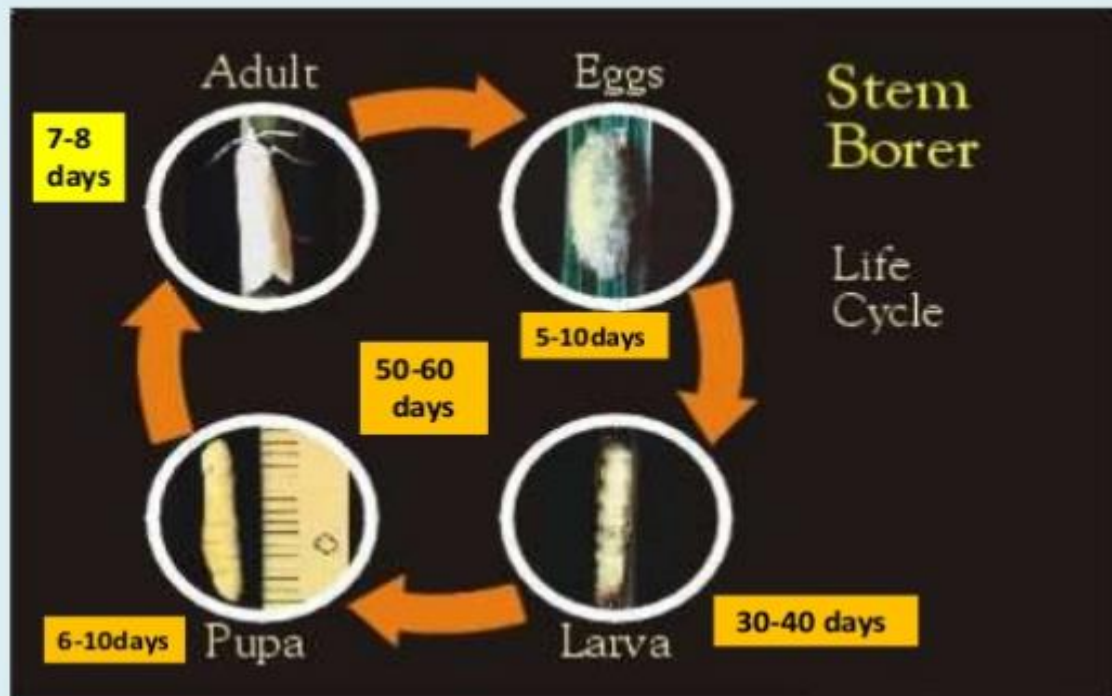
Pupa turns from yellowish-white to dark brown

Pupa turns dark brown just before adult emergence

Pupate at the base of the plant



Adult moth



Nature and symptoms of damage

Young larva, bore into the central shoots of young seedlings and tillers, and feeds internally on growing shoot causing 'dead hearts' symptom i.e drying of the central shoots and interrupting movement water and nutrients



At boot leaf stage or ear head stage – the larva bores in to the top most node and feed at the base of the ear head and succulent ear head dries and becomes white ear or chaffy ear head and such ear heads are easily come out when they are pulled out



Symptoms

- The pest can be identified with the following symptoms
- Dead heart at vegetative stage which turns brownish, curls and dries
- White ears at ear heading stage with empty, partially filled grains
- Presence of egg masses near the tips of tender leaf blades
- Activity of moths in the vicinity
- Frass at the feeding site

Reasons for attaining major pest status

- ❖ Expansion of rice area to new irrigation system
- ❖ Multiple rice cropping system
- ❖ High yielding varieties
- ❖ Application of higher dose of nitrogen
- ❖ Optimum temp.- 17- 30⁰ C for all the stage
If more than 35⁰ C are not congenial
- ❖ Warm humid condition i.e RH 80-90% are more suitable for population growth

Brown plant hopper: *Nilaparvata lugens* (Delphacidae : Hemiptera)

This is a number one enemy in Command areas , under favourable condition we can expect 100 % crop loss



Adult insects appears in two forms

Macropterous form – Fully winged form



- Have well developed wings
- Smaller abdomen
- Move from one field to another field as colonies
- Dominant at post flowering stage
- Egg laying capacity is less

Brachypterous form – Short winged form



- Have poorly developed wings
- Larger abdomen
- Can not fly, move by irrigation water
- Dominant at pre flowering stage
- Egg laying capacity is more



Young nymphs are whitish in colour and measures about 0.7mm in length and second instar nymphs are brownish black in colour and they congregate at the base of the plant and later instar are yellowish brown in colour

N.P- 15-20 days

Nature and symptoms of damage



- **Stage of attack:-** Early growth stage but some times serious infestation occurs at post flowering stage
- **Damaging stages:-** Adults (especially brachypterans) and nymphs
- **Nature of damage:-** Nymphs and adults Congregate at the base of the plant causing hopper burn
- Peak occurrence is between **Oct - Nov** and **Jan – Feb** and the pest more severe in **summer crop** due to ideal condition, the loss is heavy in high rainfall condition

Both nymphs and adults suck the sap from leaf sheath and stem at lower portion of the plant, just above the water level, continuous drying of the plants due to feeding. The crop turn yellow followed by rapid drying of plants called as **hopper burn** in concentric rings and later spread to entire plot in severe case

Management of BPH on Rice

Preventive measures

- Removal of weeds from the surrounding fields.
- Avoiding indiscriminate use of chemical insecticides as it leads to destruing of natural enemies.
- Use paddy cultivars resistant to this insect pests.
- Use light traps in the paddy field with an electric bulb as it inundates BPH insects prior to its attack on paddy
- Avoid the heavy use of Nitrogenous Fertilizers.

Biological control of Brown Plant Hopper

- Natural enemies like egg parasitoids, mired bugs, water striders and spiders prey on the BPH. Mass production of those natural enemies and release of them into the field during BPH incidence. Don't spray any insecticides after releasing of the natural enemies on to the crop.

Chemical control of Brown Plant Hopper

- Apply the insecticides most at seedling stage in the nursery seedbed.
- Spray insecticides formulations like Buprofezin, Prometrozin or other alternative insecticides.
- Avoid spraying of insecticides like chlorpyrifos, pyrethroid combinations or lambda-cyhalothrin as it favors for the resurgence and resistance of the insect pest.

Planthopper – *Nilaparvatalugens*:

There are two types of Planthopper species that attack the paddy crop. These are BPH (Brown Plant Hopper, *Nilaparvatalugens* and WBPH (White-backed Plant Hopper), *Sogatellafurcifera*.

Symptoms

- Yellowing of the plants.
- On the midrib or on leaf sheath, we can see crescent shaped white eggs that are inserted into the midrib.
- White to brown color nymphs are seen on the leaves.
- At the base of tillers, we can see brown to white adult hoppers feeding.
- At severe infestation or presence of high hopper population on the field, we can see hopper burn symptom that is leaves turn orange to yellow
- Sooty moulds are present at the base of the plant.

Major Insect of cotton

1. Latin American countries have become important suppliers of the fiber to the world market and India, Pakistan and Turkey are also the exporters. The plant is a shrub native to tropical and subtropical regions around the world, including the Americas, Africa, and India. The four cultivated species of cotton viz. *Gossypium arboreum*, *Gossypium herbaceum*, *G. hirsutum* and *G. barbadense* belong to Malvaceae family. The English term cotton derives its name from Arabic word „quṭn“, Dutch „katoen“ and French „coton“. Cotton (*Gossypium* spp), the king of fibers, usually commercial referred as white gold and one of the important commercial crops, plays a pivotal role in human civilization, economic, political and social affairs of world. Cotton is a soft, fluffy staple fiber that grows in boll, or protective capsule, around the seeds of cotton plants of the genus *Gossypium*.
2. Cotton stem: The stem can be used as organic manure or fuel. Cotton seed cake: Seed cakes contain 40% protein and serves as an important concentrated feed for livestock. Not only that, cake is the good organic matter contains 5% N, 3% P₂O₅ and 2% K₂O. Cotton seed: Depending on varieties, it contains 20-25% semi-drying edible oil (iodine no. 102) which is used for cooking. American cotton contains high oil. Fuzz: It is used in production of mattresses, surgical cotton, photographic film and paper. Cotton lint: It is the most important vegetable fiber and is woven into fabrics either alone or combined with other fibers. It is chiefly grown for its fiber which is used for manufacturing of clothes for mankind. In Nepal, textile industries also used to import raw material from neighbor countries, but due to extensive cultivation of cotton in different districts (Banke, Bardia, Dang) they get a satisfactory amount of lint from Cotton Development Board. The economic importance of cotton is based on different properties of its products
- 3.
4. A single larva can damage 30-40 bolls. Fig. Feeding injury Fig. Circular bore hole Fig. Larva and Adult Presence of granular faecal pellets outside the bore hole. Larvae seen feeding on the boll by thrusting their heads alone inside and leaving the rest of the body outside Bolls showing

regular, circular bore holes → It is one of the important major insect of cotton crop. Symptoms of damage
MAJOR INSECTS OF COTTON A list of potential insect pests of cotton with its management is listed below:
 American bollworm/Fruit borer: *Helicoverpa armiger* Family: Noctuidae
 Order: Lepidoptera

5. Adult • Light pale brownish yellow stout moth. •
6. Forewings are olive green to pale brown in colour with a dark brown circular spot in the centre. • Hind wings are pale smoky white with a broad blackish outer margin. Egg Larva Pupa → Pupa - Brown in colour, occurs in soil, leaf, pod and crop debris → It has dark brown grey lines on the body with lateral white lines and also has dark and pale bands. → Larva - Shows colour variation from greenish to brown. → Eggs - Spherical in shape and creamy white in colour, present singly → Identification of the insect
7. Releasing predator *Chrysoperla carnea* @ 1, 00, 000/ha at 6th, 13th and 14th week after sowing. → Inundative release of egg parasitoid, *Trichogramma* spp., at 6.25 cc/ha at 15 days interval 3 times from 45 DAS → Judicious water management for the crop to prevent excessive vegetative growth and larval harbourage. → Optimizing the use of nitrogenous fertilizers which will not favour the multiplication of the pest. → Removal and destruction of crop residues to avoid carry over of the pest to the next season, and avoiding extended period of crop growth by continuous irrigation. → Avoid monocropping. Growing of less preferred crops like greengram, blackgram, soyabean, castor, sorghum etc., along with the cotton as intercrop or border crop or alternate crop to reduce the pest infestation. → Avoid continuous cropping of cotton both during winter and summer seasons in the same area as well as ratooning. → Management
8. . During bolling and maturation stage, apply any one of the following insecticides (1000 l of spray fluid/ha): • Phosalone 35 EC 2.5 l/ha • Quinalphos 25 EC 2.0 l/ha • Carbaryl 50 WP 2.5 kg/ha • Pyraclofos 50 EC 1.5 l/ha • Endosulfan 35 EC 2.5 lit/ha • Cypermethrin 10 EC 600-800ml/ha
 Management contd...
 →
9. It is a major chewing insect of cotton crop. ⚡ It is a notorious pest of cotton in all cotton growing areas. ⚡ Discolored lint and burrowed seeds. Fig. Larva Fig. Adult Pink bollworm: *Pectinophora gossypiella* Family: Gelechiidae Order: Lepidoptera → The attacked buds and immature bolls drop off. →

They cut window holes (interocular burrowing) in the two adjoining seeds thereby forming "double seeds" → The holes of entry plugged by excreta of larvae which are feeding inside the seed kernels. → Rosetted flowers
→Symptoms of damage

10. Adult • Small moth. • Forewings are brown or dull yellow olive grey with dark spots on the forewing. • Hind wings margins are deeply fringed. Fig. Larva and Adult→ Larva • Shows colour variation young larva white and late instar almost black, brown or green to pale or pink • Several dark and light alternating bands running the entire length →Identification of the insect
11. Spray triazophos 40 EC 2.5l/ha and Quinalphos 20% AF in alternation even after 100 DAS.→ Release of egg parasitoids *Trichogramma chilonis* or *E. johnstoni*. → Acid delinting of cotton seeds. → Withholding irrigation water to avoid prolonged late boll production/ formation to reduce the build up of over-wintering population. → Avoid late sowing of the crop. Early sowing helps in early maturity facilitating escape. → Plough deeply to expose the hibernating larvae / pupae. → Clean cultivation and destruction of crop residues (fallen leaves, twigs etc.) before the onset of season.
→Management
12. Holes on bolls and rotting of bolls. Fig1. Drying - terminal shoots Fig2.Bore holes and rotting Fig3.Flared square→ Flaring up of bracts during square and young boll formation stage → Shedding of squares and young bolls → Drying and drooping of terminal shoots during pre – flowering stage
→ It is also a chewing insect pest of cotton that causes great economic loss to cotton crops. Symptom of damage ∞Spotted bollworms: *Earias vittella*
Spiny bollworm: *Earias insulana* Family: Noctuidae Order: Lepidoptera
13. Adult • Small buff coloured • Forewings are pea green with a wedge shaped white band running from base to out margin Fig. *E. Vitella* - Larva Fig. *E. vitella*- Adult→ Larva - Brownish with white streaks dorsally and pale yellow ventrally, Without finger shaped processes →Identification of the insect: *E. vitella*
14. Adult - Small buff coloured. Forewings are uniformly silvery green Fig. *E. insulana*-Larva Fig. *E. insulana*-Adult→ Pupa - Brown and boat shaped → Larva • Brown with dorsum showing a white median longitudinal streak. • The last two thoracic segments and all the abdominal segments have two

- pairs of fleshy tubercles (finger shaped processes) one dorsal and the other lateral → Identification of the insect : *E. insulana*
15. Spray the following insecticide • Maturity stage: -Quinalphos 25 AF 2 lit, Phosalone 35 EC 2.5 lit/ ha → Conserve and encourage the activity of spiders *Thomisus* sp., *Neosiana* sp. → Set up pheromone traps → Don't extend the crop period. → Planting trap crop of bhendi, uprooting and burning → Collect and destroy all the shed fruiting parts → Management
 16. Larvae feed the leaves by making small holes. Larva Adult → During severe attack, only the stem and side shoots will be standing in the field without any leaf or bolls → Scrapping the epidermal layer, leaving the skeleton of veins of leaf → It is a major and important chewing insect pest of cotton crop. Symptoms of damage ∞ Armyworm / Tobacco Cutworm: *Spodoptera litura* Family: Noctuidae Order: Lepidoptera.
 17. Removal and destru → Growing castor along border and irrigation bunds. → Set up the sex pheromone trap Pherodin S.L. at 12/ha → Use of light trap → Adult • Forewings – brown colour with wavy white marking • Hindwings- white colour with a brown patch along the margin Management → Larva • Pale greenish with dark markings. • Gregarious in the early stages → Egg - Laid in masses which appear golden brown → Identification of the insect Spray any one of the following insecticides • chlorpyrifos 20 EC 2.0 l/ha; • dichlorvos 76 WSC 1 lit/ha; • fenitrothion 50 EC @ 625 ml. → Hand picking and destruction of grown up caterpillars. → Collection and destruction of shed plant parts. → Removal and destruction of early stage larvae found in clusters → action of egg masses in castor and cotton crops.
 18. Use of poison bait: • Prepared with rice bran 12.5 kg, jaggery 1.25 kg, Carbaryl 50%WP 1.25 kg and water 7.5 litres. • This bait can be spread in the fields in the evening hours so that the caterpillars coming out of the soil feed and get killed. Management Contd.. → Spraying Nuclear Polyhedrosis Virus at 1.5×10^{12} POB per ha. →
 19. Development of black sooty mould due to the excretion of honey dew giving the plant a dark appearance. Aphid colony Affected leaf → Blighted appearance when infestation is severe → Stunted growth → Curling and crinkling of leaves → Infesting tender shoots and under surface of the leaves. → An important sucking insect of cotton crop. Symptom of damage ∞ Cotton aphid – *Aphis gossypii* Family: Aphididae Order Homoptera
 20. Monitoring the activities of the alate adult by setting up yellow sticky traps → Spray any one of the following insecticides (500 l spray fluid/ha) • Imidacloprid 200 SL at 100 ml/ha • Methyl demeton 25 EC 500 ml/ha •

Dimethoate 30 EC 500 ml/ha • Phosphamidon 40 SL 600 ml/ha • NSKE 5% 25 kg/ha → Release predator *Chrysoperla carnea* or *coccinella* sp. → Seed treatment with imidacloprid 70 WS at 7 g/kg protect the crop upto 8 weeks. → Adults - Greenish brown, soft bodied and small insects. • Winged forms may be seen under crowded conditions. Management → Nymphs - Yellowish or greenish brown found on the undersurface of leaves. → Identification of the insect

21. Adult - Small, slender, yellowish to brown with fringed wings • Silvery shine Fig. Adult thrip → Nymphs - Very minute, slender, yellowish and microscopic. → Silvery shine on the undersurface of leaves Identification of the insect → Attacked terminal buds – have ragged edges → Shriveling of leaves due to scrapping of epidermis and desapping → Family Thripidae Order Thysanoptera. Symptom of damage → It is a major sucking insect of cotton crop → They are active in spring. → These tiny yellow to black, selender insects are annual pest in cotton field. → Thrips: *Thrips tabaci*
22. Spray any one of the following insecticides (500 l spray fluid/ha) • Imidacloprid 200 SL at 100 ml/ha • Methyl demeton 25 EC 500 ml/ha • Dimethoate 30 EC 500 ml/ha • Phosphamidon 40 SL 600 ml/ha • NSKE 5% 25 kg/ha → Seed treatment with Imidacloprid 70 WS @ 7 g/kg protects the crop up to 8 weeks. → Management
23. It also transmits the leaf curl virus diseases of cotton. Fig. Affected leaf → Shedding of buds and bolls and poor boll opening. → Development of sooty mould . → Severe infestation results in premature defoliation → Chlorotic spots on the leaves which latter coalesce forming irregular. yellowing of leaf tissue which extends from veins to the outer edges of the leave.s → It is important sucking insect pest of cotton crop. Symptom of damage → Whitefly: *Bemisia tabaci* Family: Aleyrodidae.
24. Adult - Minute insects with yellow body covered with a white waxy bloom. Nymph Pupa Adult → Pupa - Puparia oval in shape, present on the under surface of the leaves. → Nymph - Greenish yellow, oval in outline, → Identification of the insect
25. Set up yellow pan traps and sticky traps @ 12/ ha → Cultivation of most preferred alternate host crops like brinjal, bhendi, sunflower may be avoided. → Field sanitation may be given proper attention. → Late sowing may be avoided and the crop growth should not be extended → Preferably wider spacing and judicious application of recommended dose of fertilizers → Timely sowing with recommended spacing → Removal and destruction of alternate weed hosts like *Abutilon indicum*, *Chrozophore rottlari*, *Solanum*

nigrum → Adopting crop rotation with non-preferred hosts such as sorghum, ragi, maize etc. → Growing cotton only once a year either in winter or summer season. → Management

26. Spray any one of the following plant products alone or in combination with the recommended dose of insecticide (at 2 ml/l of water) • Neem seed kernel extra → Collection and destruction of whitefly infested leaves from the plants. → Spray any one of the following in mid and late stages (1000 l spray liquid/ha) • Phosalone 35 EC at 2.5 l/ha • Quinalphos 25 EC at 2.0 l/ha • Triazophos 40 EC 2.0 l/ha • Acephate 75 SP 1.30 kg/ha Note: The use of synthetic pyrethroids should be discouraged in cotton to avoid the problem of whitefly. Cypermethrin, Fenvalrate and Deltamethrin cause resurgence of whiteflies. So avoid repeated spraying of Pyrethroids. Management contd... → Spray any one of the following in early stage (500 l of spray fluid/ha) • Methyl demeton 25 EC 500 ml or Phosphamidon 40 SL 600 ml/ha → 5% (50 kg), Neem oil at 5 ml/l of water • Fish oil rosin soap @ 25 mg /lit • Catharanthus rosea extract 5%
- 27.. Nymphs and Adults Reddish bugs with white bands on the abdomen and black markings on the wings. Fig. Red cotton bug → The bacterium *Nematospora gossypii* enters the site of injury and stains the fiber. Identification of the insect → Young bolls abort and turn dark brown → Inner boll wall with warty growth or water soaked spots → Red stained lint and rotting bolls. → Red cotton bug: *Dysdercus cingulatus* Family : Pyrrhocoridae Order : Hemiptera Symptom of damage
28. Spray Phosphamidon 100 EC@250 ml/ha → Plough the field to expose the eggs. → Management
- 29.. Nymphs and Adults - Dusky, grayish brown bug, with pointed head and hyaline wings Management Spray Phosphamidon 100 EC@250 ml/ha Fig. Dusky cotton bug → Eggs - Cigar shaped, white eggs in clusters of 2-10 within the half opened bolls, on the bolls, flower or buds → Dusky cotton bug: *Oxycarenus hyalinipennis* Symptom of damage Sucks the sap from developing seeds in open bolls and stains the lint black. Seeds discolored and shrunken. Identification of the insect
30. Spray application of any following insecticides viz. • Carbaryl 50 WP @ 1kg/acre, • Thiodicarb 75 WP @ 250 g/acre • Profenophos 50 EC @ 500 ml/acre • Acephate 75SP @ 800 g/acre. → Affected plants appear sick and black, resulting reduced fruiting capacity. Management → Excrete copious amount of honey dew on which the fungus sooty mould grow. → Heavy clustering of mealy bugs usually seen under surface of leaves as a thick mat with waxy secretion. → They are considered pests as they feed on plant

juices and act as vector of various diseases. Symptom of damage ☞ Mealy bugs: *Phenacoccus* sp., *Ferrisa* sp. and *Maconellicoccus* sp. Family: *Psudococcidae* Order: *Hemiptera*

31. Older plants that survive, lack vigor and strength, and when strong winds blow, these plants sometimes break at the nodes. Figs. Showing symptoms → Young plants are invariably killed → Swellings on the stem just above the ground level. → It is a serious pest of cotton crop. Symptoms of damage ☞ Cotton Stem Weevil: *Pempherus* (*Pempherulus*) *affinis* Family: *Curculionidae* Order: *Coleoptera*
32. Basal application of FYM 25 t/ha or 250 kg/ha of Neem cake. → Soil application of Carbofuran 3 G @ 30 kg may be done on 20 days after sowing and earthed up. → Adult - Weevil dark greyish brown with pale cross bands on the elytra. Management → Bore hole surrounded by raised margins Identification of the insect → Terminal shoots with galls → Shoot weevil: *Alcidodes affaber* Family: *Curculionidae* Order: *Coleoptera*. Symptoms of damage
33. Basal application of FYM 25 t/ha or 250 kg/ha of Neem cake. → Soil application of Carbofuran 3 G @ 30 kg may be done on 20 days after sowing and earthed up. → Adult - Dark brown jewel beetle Management → Wilting in patches Identification of the insect → Plants with drooping leaves, → Stem borer: *Sphegnoptera gossypii* Symptoms of damage
34. Plants defoliated in severe attack Fig. Leaf affected by leaf roller → Marginal portion of leaves eaten away → Leaves rolled in the form of trumpets fastened by silken threads → It is a important insect which cause rolling of leaves of cotton. Symptom of damage ☞ Leaf roller: *Sylepta derogata* Family: *Pyrsultidae* Order: *Lepidoptera*.
35. Spray any one of the following insecticides : • chlorpyrifos 20 EC 2.0 l/ha • dichlorvos 76 WSC 1 lit/ha • fenitrothion 50 EC @ 625 m → Hand picking and destruction of grown up caterpillars. → Collection and destruction of shedded plant parts. → Adult - Moth with yellow wings having brown wavy markings Fig. Larva Fig. Adult Management → Larva - Bright green (glistening) with dark head and prothoracic shield. → Identification of the insect l.
36. The leaves dried up and are shed and the growth of the crop is retarded. Identification of the insect Nymph - Light green, translucent, wingless found between the veins of leaves on the under surface Adult - Green, wedge shaped leafhopper. Fig. Leafhopper → The margin of the leaves get broken and crumble into pieces when crushed. → In the case of severe infestation,

leaves get a bronze or brick red colour which is typical “hopper burn” symptom. → The margin of the leaves start curling downwards and reddening sets in. → Tender leaves become yellow. → Leafhopper: *Amrasca biguttula biguttula* devastans Family: Cicadellidae Order: Hemiptera. Symptoms of damage

37. Spray any one of the following insecticides (500 l spray fluid/ha) • Imidacloprid 200 SL at 100 ml/ha • Methyl demeton 25 EC 500 ml/ha • Dimethoate 30 EC 500 ml/ha • Phosphamidon 40 SL 600 ml/ha • NSKE 5% 25 kg/ha → Where the leafhopper is a big menace, apply neem oil formulation 0.5 % or neem oil 0.5 % thrice at fortnightly intervals → Early sowing and closer spacing of cotton reduces pest infestation → Seed treatment with Imidacloprid 70 WS at 7 g/kg protect the crop upto 8 weeks. → Management

38. Ants are an important predator of bollworm egg and larvae. Fig. Ant → Several different species may inhabit fields as the worker caste ants (approximately 1/8 inch in length, wingless and with a narrow waist) → Ants are important and overlooked predators in cotton fields. → Some of the beneficial insects are as follows: Ants → These beneficial species attack the egg, immature (larval; nymphal and pupal) and stage of most insect species. → Besides harmful insects numerous beneficial insects and arthropod species can be found in cotton fields attacking pest species. → SOME USEFUL INSECTS OF COTTON

39. Damsel bugs also attacks spider mites and bollworms (egg, small to medium sized larvae). Fig. Assassin bug Fig. Big-eyed bug → Minute pirate bugs attack Thrips, aphids, mites and bollworm (eggs, larvae). Both big-eyed bug adults and nymphs also frequently attack thrips, plant bugs, bollworm and other insects. → It readily attacks the larval stage of Bollworm. → Several different species of Assassin bugs (yellow to black coloration) are predators of both immature and adult stages of insects. → Assassin bugs and other piercing-sucking predators

40. A single lacewing larva can consume more than 40 bollworm egg in one day. Fig. Lacewings Larva Fig. Lacewings-adult → Adult may feed on insects’ eggs, but the larvae are more important because they are voracious predators of aphids and bollworms (eggs, larvae). → Two lacewing species (brown and green) occur in cotton fields, but the green is more commonly found. → Lacewings

Major pest of Groundnut

Major pests

1. **Aphids**
2. **Leaf hopper**
3. **Thrips**
4. **Red hairy caterpillar**
5. **Leaf miner**
6. **Tobacco caterpillar**
7. **Gram pod borer**
8. **Pod borer**
9. **White grub**
10. **Termites**

Oilseeds occupy a prominent place among the principal commercial crops grown in India. The important oilseeds cultivated in India are *Brassica sp*, groundnut, sunflower, safflower, castor, sesame and linseed. These crops are damaged by number of pests, of which mustard aphid, mustard sawfly and the painted bug are the most serious. The aphid is the most serious pest on brassica oilseeds throughout India.

1. Aphids

Aphis craccivora

(Aphididae: Hemiptera)

- **Distribution and status:**

India, Africa, Argentina, Chile, U.S.A. Europe and Australia.

- **Host Plants:**

Groundnut, beans, safflower, lablab, Niger, peas, pulses and some weeds.



Life Cycle

- The pest breeds **parthenogenetically** and They grow very fast and are full-fed in 7-10 days.
- A Female Aphid Lives for 9 to 25 Days And Can Produce From **25 to 125 Young** during its life.
- There may be up to 20 generations in the years. By november winged forms have developed and eggs are laid before winter sets in.

Damage symptoms

- Both **nymphs and adults** suck the sap from the leaflets and tender shoots mostly up to two months after germination.
- It results in **wilting of tender shoots** during hot weather. Leaves mottled with **chlorotic** or dark green spots and plant growth becomes stunted.
- Sometimes **honey dew** deposited on the leaves and shoots could be seen which attract the ants.



IPM STRATEGIES

Cultural Control

- **Timely sowing** of the crop.
- Varieties which are densely hairy and with stiff leaves interfere Pest population.

Mechanical Control

- **Hand picking** and destruction of various insect stages and the affected plant parts.

Biological Control

- Apply **NSKE 5%** (neem seed kernel extract) to control sucking pests.
- Augment the release of *Cheilonenes sexmaculata* @ 1250/ha.
- Conserve bio agents like flower bugs (anthocorids), lady bird beetles (coccinellids), praying mantis, hover flies (syrphids), **green lace wing** (chrysopids), long horned grass hoppers and spiders.

Chemical Control

- Apply safe chemical insecticides at recommended doses only if the insect population crosses the ETL.
- Young crop may be applied with **monocrotophos 36SL** 10 ml in 10 lit. of water or **Dimethoate 30 EC** @ 10 ml in 10 lit. of water.

2. Leaf hopper

Empoasca kerri

(Cicadellidae: Hemiptera)

Host Plants

Brinjal, Sesamum, Cotton, Castor, Pulses, Vegetables etc.



Identification And Life History

- Elongate, active wedge shaped green insects found on the under surface of leaves.
- The eggs are inserted in to the leaf tissue close to the midrib or into the petiole.
- The egg hatch in a week and nymphs into adult in 10 days.
- The infestation is high during August and September and February and March.

Damage symptoms

Both adults and nymphs suck sap

from young leaves mostly from the lower surface.

The first symptom of attack is a whitening of the veins. Chlorotic (yellow) patches then appear, especially at the tips of leaflets, probably caused by a reaction between the jassids salivary secretion and plant sap



Under severe infestation, the leaf tips become necrotic in a typical V shape giving the crop a scorched appearance known as 'hopper burn'.

IPM STRATEGIES

Cultural Control

- Timely sowing of the crop and field sanitation.
- Grow tolerant varieties And Crop rotation with non-host crop.
- Intercropping with pearl millet, Avoid groundnut-castor inter crop, it increases the infestation.
- Irrigate once to avoid prolonged mid-season drought to prevent pre-harvest.

Mechanical Control

- Collect and destroy the affected parts of the plant

Biological Control

- Conserve bio agents like **praying mantis**, long horned **grass hoppers**, **dragon flies**, **spiders**, green muscardine fungus.

Chemical Control

- Apply safe chemical insecticides at recommended doses only if the insect population crosses the ETL.
- Spray **dimethoate 30EC** @ 10 ml in 10 lit. of water or monocrotophos 36SL @ 10 ml in 10 liter water.
- **Imidachloprid 17.8 SL** 3 ml in 10 lit. of water

Coconut Pests

- **Rhinoceros beetle:** *Oryctes rhinoceros* Guest (Coleoptera: Scarabaeidae)
- **Red palm weevil:** *Rhynchophorus ferrugineus* Olivier (Coleoptera: Curculionidae)
- **Black headed caterpillar:** *Opisina arenosella* Walker (Lepidoptera: Xylorictidae)
- **Eriophyid mite:** *Aceria guerreronis* Keifer (Acarina: Eriophyidae)
- **Termite:** *Odontotermes* sp (Isoptera: Termitidae)

Diseases

- **Bud rot:** *Phytophthora palmivora* Butler
- **Stem bleeding:** *Thielaviopsis paradoxa* De Seynes
- **Leaf rot:** *Exserohilum rostratum* & *Colletotrichum gleosporioides* Drechsler
- **Cadang cadang disease:** Coconut cadang-cadang viroid (CCCVd)

II. M.Sc., ZOOLOGY

Elective paper IV. Applied Entomology

Dr. M.MUTHUSAMY. Associate professor

UNIT:IV

Management

- Cultural Control
 - Pick trees suited to location
 - Avoid excess nitrogen
 - Add supplemental water during drought
 - Avoid stock-piling cut lumber
 - Avoid injuring trees
 - Look for resistant varieties

Management

- Physical Control
 - washing foliage
 - Barriers such as sand, salt, copper shields.



Natural Enemies

- Parasites
- Pathogens
- Predators
- Birds
- Lizards



Insect/Pest Control Methods

➤ Insect:

Any organism which has three pairs of legs(6 legs) and body divided into three parts i-e "Head, Thorax & Abdomen".

➤ Pest:

Any organism that damages crops, injures or irritates livestock or man, or reduces the fertility of land (Broader term).

➤ Control:

The killing, reduction or elimination of insects which are pests, using insecticide, etc to avoid damage.

➤ Methods:

- The methods should be most effective & less harmful to environment(Humans & animals).
- Control methods includes everything that makes life hard for insects.
- Control methods also tend to kill insects to prevent their increase in particular area.
- Insects are kept under control in two ways:

I. NATURAL CONTROL:

- The measures that destroy or check insects without the influences of the man.
- This includes: Climatic, topographic & biotic factors.

Be the first to clip this slide

a) Climatic factors:

- Include rainfall, cold, heat and wind.
- These climatic factors effect the insect population to great extent.
- Rainfall and hail storm kill insect in larval stage.
- Extreme heat and cold temperature affect a number of insects spp.
- Wind movement is also of great importance: i-e many spp of smaller insects which normally fly for considerable distance are beaten by the wind.

b) Topographic factors:

- Include rivers, lakes, mountains, types of soil & other barriers.
- Rivers and lakes affect those insects which do not possess the power of flight.
- Mountains offer varying conditions of climatic through which many insects cannot pass.
- The character of soil of any region affects a marked influence on the insects.
- For example: Certain spp of tiger beetle larvae live in sandy soil and are unable to exist in the clay soil.

c) Biotic factors:

- Include predatory insects, birds, reptiles, mammals, parasitic insects, fungi, bacteria & other microorganisms.
- A large number of insects parasite on other insects.
- Some are predators of insects.
- Almost all birds feed on adult insects as well as other worms.
- Insects are also attacked by fungi, bacterial & viral diseases.

II) APPLIED CONTROL:

- To destroy insects by the uses of man made methods.
- This includes: Silvicultural, biological, mechanical, physical & chemical control methods.
- Another method is "Integrated pest management(IPM)" is new concept of insect control.
- IPM minimizes harm to the environment while keeping problem insects in check.
- Silviculture methods include the practices which destroy insects i-e mixed cropping & seedling etc.
- Biological method is the control of insects by other living i-e predation.
- Mechanical method is the control of insects through mechanical means i-e collection of insects by hand.
- Physical method is the control of insects by the increase or decrease in temperature, humidity & wind.
- Chemical method is the control of insects through some chemicals i-e insecticides, herbicides, pesticides etc.
- ❑ Other methods of insect control include:

a. Raising healthy nurseries:

- Diseased free seeds should be collected from healthy trees & raised in nurseries.
- Excellent preparation of seeds beds and proper irrigation & fertilizing in the nurseries should be ensured from healthy nurseries

b. Mixed forest:

- Mixed forests prevent the spread of specific insects.
- For example; Shisham defoliator can be controlled by planting it with mulberry.

Mechanical

- Pests are controlled by non-chemical direct measures,
- Examples include:
 1. *hand picking to remove insects*
 2. *tilling to remove weeds*
 3. *trapping to catch insects or rodents*